

Erdős Problem #643

EP-643

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $f(n; t)$ be minimal such that if a t -uniform hypergraph on n vertices contains at least $f(n; t)$ edges then there must be four edges A, B, C, D such that $A \cup B = C \cup D$ and $A \cap B = C \cap D = \emptyset$. Estimate $f(n; t)$ - in particular, is it true that for $t \geq 3$ $f(n; t) = (1 + o(1))\binom{n}{t-1}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>